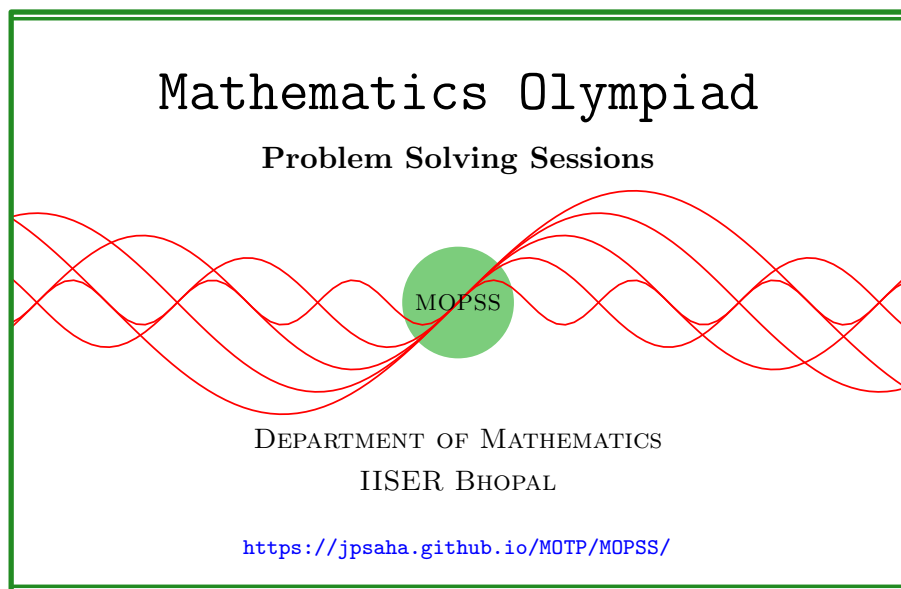


Integer divisibility

MOPSS



Suggested readings

- Evan Chen's advice [On reading solutions](https://blog.evanchen.cc/2017/03/06/on-reading-solutions/), available at <https://blog.evanchen.cc/2017/03/06/on-reading-solutions/>.
- Evan Chen's [Advice for writing proofs/Remarks on English](https://web.evanchen.cc/handouts/english/english.pdf), available at <https://web.evanchen.cc/handouts/english/english.pdf>.
- [Notes on proofs](#) by Evan Chen from [OTIS Excerpts](#) [[Che25](#), Chapter 1].
- [Tips for writing up solutions](https://www.math.utoronto.ca/barbeau/writingup.pdf) by Edward Barbeau, available at <https://www.math.utoronto.ca/barbeau/writingup.pdf>.
- Evan Chen discusses why [math olympiads](#) are a valuable experience for [high schoolers](#) in the post on [Lessons from math olympiads](#), available at <https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/>.

List of problems and examples

1.1	Example	2
1.2	Example	2
1.3	Exercise (IOQM 2023 P12, AoPS)	2
1.4	Example (India RMO 2016g P8)	4
1.5	Example	4
1.6	Exercise (USAMO 1974 P1, AoPS)	5
1.7	Example	5
1.8	Exercise (Tournament of Towns Spring 2014, Senior A Level P4, by G.K. Zhukov)	5
1.9	Exercise (IMO 2006 P5, AoPS, by Dan Schwarz, Romania) .	7

§1 Integer divisibility

Example 1.1. If P is a polynomial with integer coefficients and a, b are integers, then $P(a) - P(b)$ is a multiple of $a - b$.

Walkthrough —

(a) Show that it suffices to prove it for monomials.

Solution 1. ■

Example 1.2. Let $P(x)$ be a polynomial with integer coefficients such that $P(0), P(1)$ are odd. Show that $P(x)$ does not have any integer root.

Solution 2. If $P(x)$ admits an integer root α , then $\alpha, \alpha - 1$ are odd, which is impossible. ■

Exercise 1.3 (IOQM 2023 P12, AoPS). Let $P(x) = x^3 + ax^2 + bx + c$ be a polynomial where a, b, c are integers and c is odd. Let p_i be the value of $P(x)$ at $x = i$. Given that $p_1^3 + p_2^3 + p_3^3 = 3p_1p_2p_3$, find the value of $p_2 + 2p_1 - 3p_0$.

Summary — Using parity conditions, show that $p_1 + p_2 + p_3 \neq 0$. This gives $p_1 = p_2 = p_3$. This gives a system of equations in a and b , which can be solved to find a and b .

Walkthrough —

(a) Note that if $p_1 + p_2 + p_3 \neq 0$, then using the condition that p_1, p_2, p_3 are real, it follows that $p_1 = p_2 = p_3$.

(b) Note that

$$\begin{aligned}p_1 &= 1 + a + b + c, \\p_2 &= 8 + 4a + 2b + c, \\p_3 &= 27 + 9a + 3b + c,\end{aligned}$$

which gives that

$$p_1 + p_2 + p_3 = 36 + 14a + 6b + 3c.$$

Since c is an odd integer, it follows that^a $p_1 + p_2 + p_3 \neq 0$. Consequently, $p_1 = p_2 = p_3$.

(c) This gives

$$3a + b = -7, 5a + b = -19,$$

and hence, $a = -6, b = 11$.

(d) This gives

$$\begin{aligned}p_2 + 2p_1 - 3p_0 &= 10 + 6a + 4b + 3c - 3c \\&= 18.\end{aligned}$$

^aSince c is odd, it follows that p_2 is odd. Note that $p_1 - p_3$ is even (Why? Does it follow that $f(1) - f(3)$ is divisible by 2 for any polynomial $f(x)$ with integer coefficients?), and hence, $p_1 + p_2 + p_3$ is odd.

Solution 3. Since $P(x)$ is a polynomial with integer coefficients, it follows that $p_1 - p_3$ is divisible by 2. Since c is odd, it follows that p_2 is odd. This shows that $p_1 + p_2 + p_3$ is odd, and hence, $p_1 + p_2 + p_3 \neq 0$. Using $p_1^3 + p_2^3 + p_3^3 = 3p_1p_2p_3$, we obtain

$$(p_1 + p_2 + p_3)(p_1^2 + p_2^2 + p_3^2 - p_1p_2 - p_2p_3 - p_3p_1) = 0.$$

This gives

$$p_1^2 + p_2^2 + p_3^2 - p_1p_2 - p_2p_3 - p_3p_1 = 0,$$

or equivalently,

$$(p_1 - p_2)^2 + (p_2 - p_3)^2 + (p_3 - p_1)^2 = 0,$$

which implies that $p_1 = p_2 = p_3$. This shows that

$$1 + a + b + c = 8 + 4a + 2b + c = 27 + 9a + 3b + c,$$

which yields

$$3a + b = -7, 5a + b = -19.$$

This gives $a = -6, b = 11$. It follows that

$$p_2 + 2p_1 - 3p_0 = 10 + 6a + 4b + 3c - 3c$$

$$\begin{aligned}
 &= 10 - 36 + 44 \\
 &= 18.
 \end{aligned}$$

■

Example 1.4 (India RMO 2016g P8). At some integer points a polynomial with integer coefficients take values 1, 2 and 3. Prove that there exist not more than one integer at which the polynomial is equal to 5.

Solution 4. Denote the polynomial by $P(x)$. On the contrary, let us assume that there are at least two distinct integers where $P(x)$ takes the value 5.

Let a, b, c be integers such that

$$P(a) = 1, P(b) = 2, P(c) = 3.$$

Note that $a - b$ divides $P(a) - P(b)$, $b - c$ divides $P(b) - P(c)$. It follows that $a - b = \pm 1, b - c = \pm 1$. Since a, b are of opposite parity, and so are the integers b, c , we obtain that a, c are of the same parity. Noting that $c - a$ divides $P(c) - P(a) = 2$, it follows that $c - a = \pm 2$. Combining this with $a - b = \pm 1, b - c = \pm 1$, we get $a - b = b - c = 1$ or $a - b = b - c = -1$.

This shows that $P(b - 1) = 1, P(b) = 2, P(b + 1) = 3$ holds or $P(b + 1) = 1, P(b) = 2, P(b - 1) = 3$ holds. Note that in the first case, the polynomial $R(x) := P(x - b)$ takes the values 1, 2, 3 at the integers $-1, 0, 1$ respectively. In the second case, the polynomial $S(x) = P(-x + b)$ takes the values 1, 2, 3 at the integers $-1, 0, 1$ respectively. This proves that there is a polynomial $Q(x)$ with integer coefficients which takes the values 1, 2, 3 at $-1, 0, 1$ respectively.

From the hypothesis, it follows that there are distinct integers i, j such that $Q(i) = Q(j) = 5$. Note that $i - 1$ divides $Q(i) - Q(1) = 2$, i divides $Q(i) - Q(0) = 3$, $i + 1$ divides $Q(i) - Q(-1) = 4$. Since i divides 3, we obtain $i = \pm 1, \pm 3$. Using $Q(-1) = 1, Q(1) = 3$, we get $i \neq -1, i \neq 1$. This gives $i = \pm 3$. Noting that $i - 1$ divides 2, we obtain $i \neq -3$, and hence $i = 3$. Similarly, it follows that $j = 3$. ■

Example 1.5. Let $P(x)$ be a polynomial with integer coefficients such that $P(20), P(25)$ are of absolute value equal to 1. Show that $P(x)$ does not vanish at any integer.

Walkthrough —

- (a) If $P(x)$ vanishes at an integer α , then $\alpha - 20$ divides $P(20)$ and $\alpha - 25$ divides $P(25)$.

Solution 5. On the contrary, let us assume that $P(x)$ vanishes at an integer α . Note that $\alpha - 20$ divides 1, and so does $\alpha - 25$. This shows that $\alpha - 20, \alpha - 25$ are of absolute value equal to 1. Applying triangle inequality, we obtain

$$5 \leq |\alpha - 20| + |\alpha - 25| \leq 2,$$

which is impossible. ■

Exercise 1.6 (USAMO 1974 P1, AoPS). Let a , b , and c denote three distinct integers, and let P denote a polynomial having all integral coefficients. Show that it is impossible that $P(a) = b$, $P(b) = c$, and $P(c) = a$.

Walkthrough —

- (a) Show that each of the integers $a - b$, $b - c$, $c - a$ is a multiple of the remaining two integers.
- (b) Prove that this implies that a, b, c are equal.

Solution 6. Note that

$$a - b \mid P(a) - P(b) = b - c \mid P(b) - P(c) \mid c - a \mid P(c) - P(a) = a - b.$$

Consequently, the integers $a - b$, $b - c$, $c - a$ are of the same absolute value. Denote their absolute value by k . Note that their sum is zero. However, the sum is equal to mk , for some $m \in \{\pm 1, \pm 3\}$. Hence, k is equal to zero. This yields that $a = b = c$. ■

Here is a more general result.

Example 1.7. Let $P(x)$ be a polynomial with integer coefficients, and let n be an odd positive integer. Suppose that $x_1, x_2, \dots, x_n$ is a sequence of integers such that $x_2 = P(x_1)$, $x_3 = P(x_2)$, $\dots$, $x_n = P(x_{n-1})$, and $x_1 = P(x_n)$. Prove that all the x_i 's are equal.

Walkthrough — Show that

$$a_1 - a_2 \mid a_2 - a_3 \mid a_3 - a_4 \mid \dots \mid a_n - a_1 \mid a_1 - a_2.$$

Note that sum of these differences is an odd multiple of their absolute value.

Exercise 1.8 (Tournament of Towns Spring 2014, Senior A Level P4, by G.K. Zhukov). In the plane, the points with integer coordinates (x, y) satisfying $0 \leq y \leq 10$ are marked. Consider a polynomial of degree 20 with integer coefficients. Determine the maximum possible number of marked points which can lie on its graph.

Walkthrough —

- (a)

Solution 7. Note that the polynomial

$$(x-1)(x-2)(x-3)\dots(x-20)$$

of degree 20 has integer roots. Let us prove that the graph of no polynomial of degree 20 with integer coefficients passes through more than 20 marked points.

Claim — Let $P(x)$ be a polynomial of degree 20 with integer coefficients. No more than 20 marked points lie on the graph of $P(x)$.

Proof of the Claim. On the contrary, let us assume that there are integers $x_1 < x_2 < \dots < x_{21}$ such that

$$0 \leq P(x_i) \leq 10$$

holds for all $1 \leq i \leq 21$. For any integer $1 \leq i \leq 10$, the inequality $x_{21} - x_i \geq 11$ holds, and using that $x_{21} - x_i$ divides the integer $P(x_{21}) - P(x_i)$, which lies in $[-10, 10]$, it follows that $P(x_{21}) = P(x_i)$. Similarly, for any integer $12 \leq i \leq 21$, it follows that $P(x_1) = P(x_i)$. This shows that $P(x_i) = P(x_1)$ for any integer $i \in \{1, 2, \dots, 10\} \cup \{12, 13, \dots, 21\}$. Since $P(x)$ is a polynomial of degree 20, it follows that

$$P(x) - P(x_1) = c(x - x_1)(x - x_2)\dots(x - x_{10})(x - x_{12})(x - x_{13})\dots(x - x_{21})$$

holds for some nonzero integer a . This yields that

$$|P(x_{11}) - P(x_1)| \geq (10!)^2,$$

which is impossible. □

This proves that the maximum possible number of marked points which can lie on the graph of a polynomial of degree 20 with integer coefficients is equal to 20. ■

Lemma 1

Let P be a polynomial with integer coefficients. Suppose a is an integer and k is a positive integer such that $P^k(a) = a$, where P^k denotes the k -fold composite map from $\mathbb{Z} \rightarrow \mathbb{Z}$. Show that $P^2(a) = a$.

Proof. Let ℓ denote the smallest positive integer such that $P^\ell(a) = a$. If $\ell = 1$ or $\ell = 2$, then we are done. Henceforth, we assume that $\ell \geq 3$.

Note that

$$P(a) - a \mid P^2(a) - P(a) \mid \dots \mid P^\ell(a) - P^{\ell-1}(a) = a - P^{\ell-1}(a) \mid P(a) - a.$$

Since $a - P^{\ell-1}(a)$ is nonzero, it follows that the above differences are nonzero. Consequently, for any $1 \leq i \leq \ell$,

$$P^{i+1}(a) - P^i(a) = \pm(P^i(a) - P^{i-1}(a)).$$

If $P^{i+1}(a) = P^{i-1}(a)$ holds for some $1 \leq i \leq \ell$, then applying $P^{\ell-i+1}$ to both sides, we obtain $P^2(a) = a$, which contradicts the assumption that $\ell \geq 3$. It follows that for any $1 \leq i \leq \ell$,

$$P^{i+1}(a) - P^i(a) = P(a) - a$$

holds, which implies that

$$\sum_{i=0}^{\ell-1} (P^{i+1}(a) - P^i(a)) = \ell(P(a) - a).$$

This gives $P(a) = a$, which contradicts the assumption that $\ell \geq 3$. This completes the proof. $\square$

Exercise 1.9 (IMO 2006 P5, AoPS, by Dan Schwarz, Romania). Let $P(x)$ be a polynomial of degree $n > 1$ with integer coefficients, and let k be a positive integer. Consider the polynomial $Q(x) = P(P(\dots P(P(x)) \dots))$, where P occurs k times. Prove that there are at most n integers t such that $Q(t) = t$.

Walkthrough —

(a) Does the above lemma help?

Solution 8. By the above lemma, it reduces to considering the case $Q(x) = P^2(x)$.

Suppose Q has more than n fixed points. Since P is not linear, it follows that P cannot have n fixed points, and hence not all the fixed points of Q are fixed points of P . Let b be a non-fixed point of P , and $Q(b) = b$. Suppose a be a fixed point of Q , other than b .

Let us first consider the case that $P(a) \neq a$. Note that

$$P(b) - a \mid P(a) - b \mid P(b) - a$$

holds, and

$$a - b \mid P(a) - P(b) \mid a - b$$

holds too. This yields that

$$|P(a) - b| = |P(b) - a|, \quad |P(a) - P(b)| = |a - b|.$$

If

$$P(a) - b = a - P(b), \text{ and } P(a) - P(b) = a - b$$

hold, then b would be a fixed point of P . It follows that at least one of

$$P(a) - b = -(a - P(b)), P(a) - P(b) = -(a - b)$$

holds. Consequently, we obtain

$$P(a) + a = P(b) + b.$$

Next, let us consider the case that $P(a) = a$. Note that

$$P(b) - a \mid b - a \mid P(b) - a.$$

Since b is not a fixed point for P , it follows that

$$P(b) - a = a - b,$$

which yields

$$P(a) + a = P(b) + b.$$

This proves that all the roots of $Q(x) = x$ are the roots of $P(x) + x = P(b) + b$. Since $P(x)$ has degree $n > 1$, it follows that the polynomial $P(x) + x - P(b) - b$ is of degree n , and it has more than n roots, which is impossible.

Hence, there are at most n integers t such that $Q(t) = t$ holds. ■

References

- [Che25] EVAN CHEN. *The OTIS Excerpts*. Available at <https://web.evanchen.cc/excerpts.html>. 2025, pp. vi+289 (cited p. 1)